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Marketing Analytics Portfolio Project

Marketing ROI & Budget Reallocation Strategy

Marketing Analytics Case Study

Project Objective
Develop a data-driven marketing optimization framework to improve ROI using attribution analysis, regression modeling, and budget reallocation strategy.

Project Scope

- Marketing performance diagnosis
- Attribution vs Incrementality analysis
- Revenue impact modeling
- Budget optimization
- Revenue forecasting
- Executive dashboard development

Analytical Methods

Data Engineering • Attribution Modeling • Regression Analysis • Forecasting • Dashboarding

February 2026

Executive Summary

This project evaluates marketing performance and identifies opportunities to improve ROI through data-driven budget optimization.

Revenue	**Rs.102.1M**
Marketing Spend	**Rs.74.4M**
ROAS	**1.37**
Orders	**16.2K**
Model Accuracy	**58%**
Recommended Budget	**Rs.10.6M**
Projected Revenue	**Rs.7.7M**

Marketing efficiency differences across channels indicate strong opportunity for strategic budget optimization.

Key Insights

- Organic and Email channels deliver strongest efficiency
- Search drives highest revenue scale
- Paid Social shows weakest ROI despite high spend
- Returning customers generate majority revenue
- Marketing explains 58% of revenue movement

Recommendation
Prioritize high efficiency channels in future budget planning while reducing inefficient campaigns.

Worst Case Revenue	**Rs.6.9M**
Base Case Revenue	**Rs.7.7M**
Best Case Revenue	**Rs.8.4M**
Estimated Margin Improvement	**35%**

Executive Conclusion

The analysis indicates that marketing efficiency varies significantly across channels, creating an opportunity to improve ROI through targeted reallocation. Prioritizing high-efficiency channels while reducing inefficient spend is expected to improve revenue productivity without increasing total marketing investment.

Business Takeaways

- Marketing ROI varies significantly across channels indicating optimization potential
- Revenue concentration among returning users suggests strong retention economics
- Regression results confirm marketing has measurable revenue impact
- Budget optimization shows opportunity to improve efficiency without increasing spend

Performance Diagnosis

Revenue trends indicate stable growth supported by marketing activity

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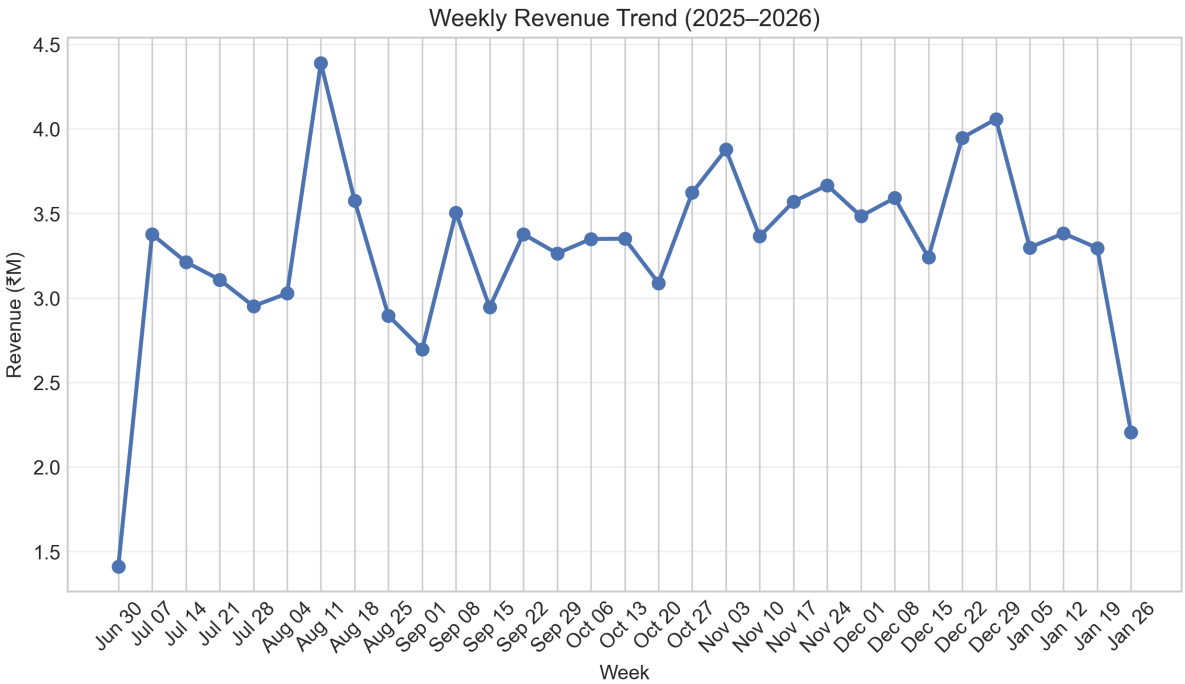

Figure 1: **Exhibit 1 — Revenue trends show stable growth with seasonal spikes**

> Revenue growth correlates with marketing efficiency improvements indicating strong campaign effectiveness.

This suggests marketing investments are contributing to revenue stability rather than volatile growth patterns.

Key Observations

- Revenue remains stable with periodic growth spikes
- Marketing activity appears to support baseline revenue stability
- No extreme volatility suggests predictable demand patterns

Marketing Channel Efficiency Diagnosis

Channel ROI differences indicate budget reallocation opportunity

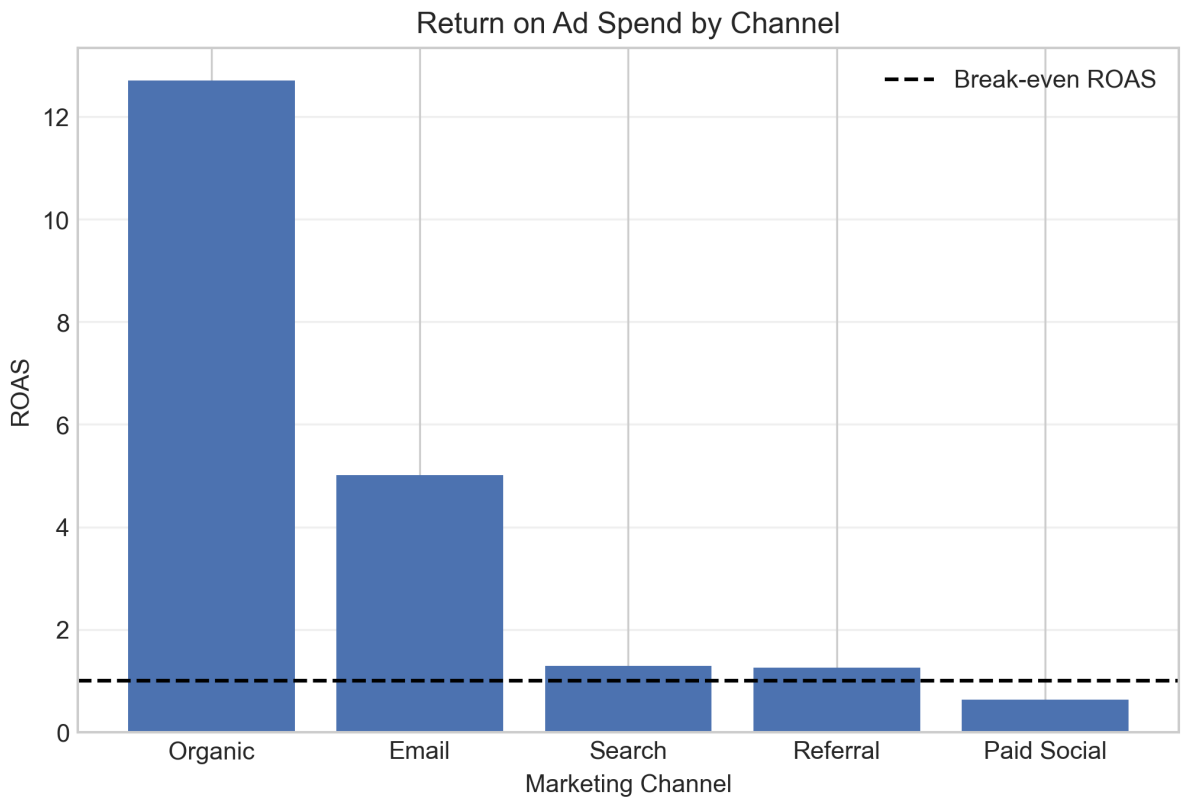

Figure 2: **Exhibit 2: Organic and Email deliver highest ROI while Paid Social underperforms**

Key Findings

- Organic shows highest ROAS
- Email shows strong conversion efficiency
- Search drives largest revenue scale
- Paid Social shows weakest efficiency

These differences suggest significant opportunity to improve efficiency by reallocating spend from lower ROI channels toward higher performing channels.

Strategic Interpretation

- High ROAS channels indicate scalable growth opportunities
- Low efficiency channels suggest optimization or restructuring need
- Balanced mix required between scale and efficiency

Customer and Segment Insights

Revenue concentration among returning users highlights retention importance

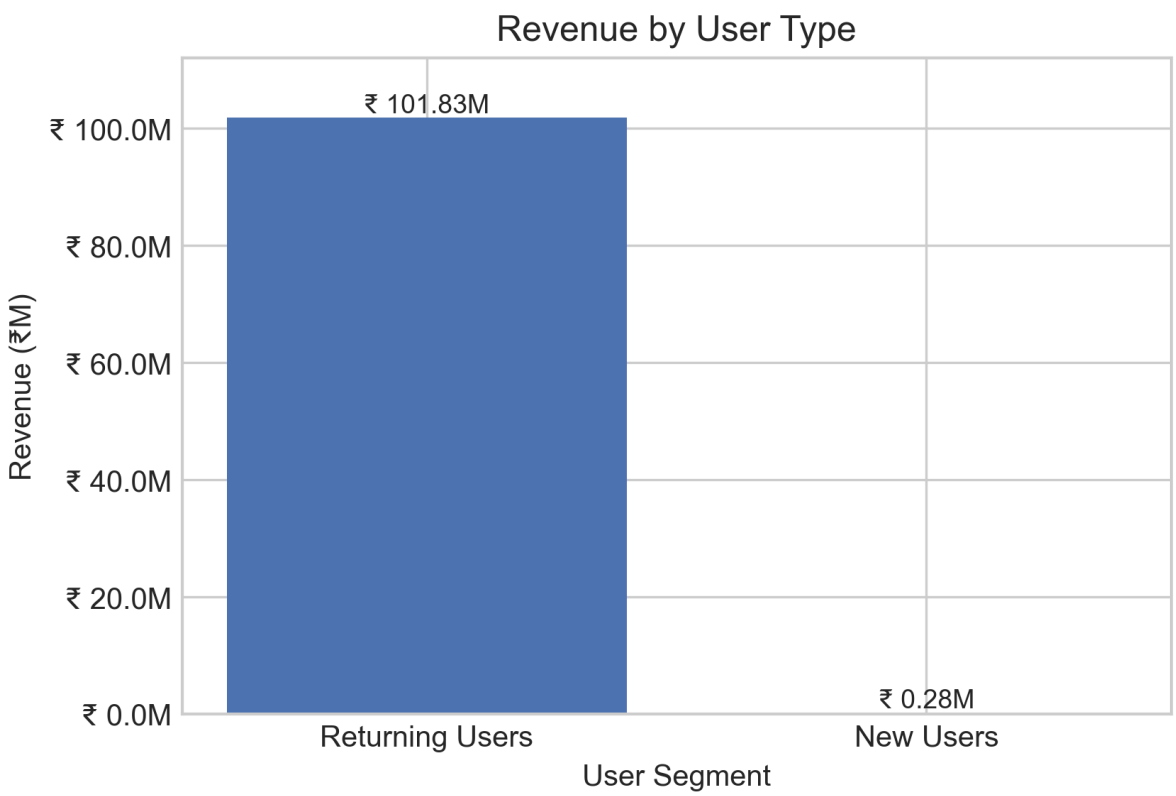

Figure 3: **Exhibit 3 — Returning customers generate majority revenue**

Returning users represent strongest revenue source indicating strong retention economics.

This highlights the importance of retention marketing and lifecycle engagement strategies over pure acquisition growth.

Segment Implications

- Customer retention likely more profitable than acquisition expansion
- Returning users may respond well to loyalty campaigns
- Opportunity exists for lifecycle marketing improvements

Attribution vs Incrementality Analysis

Regression reveals incremental impact beyond attribution

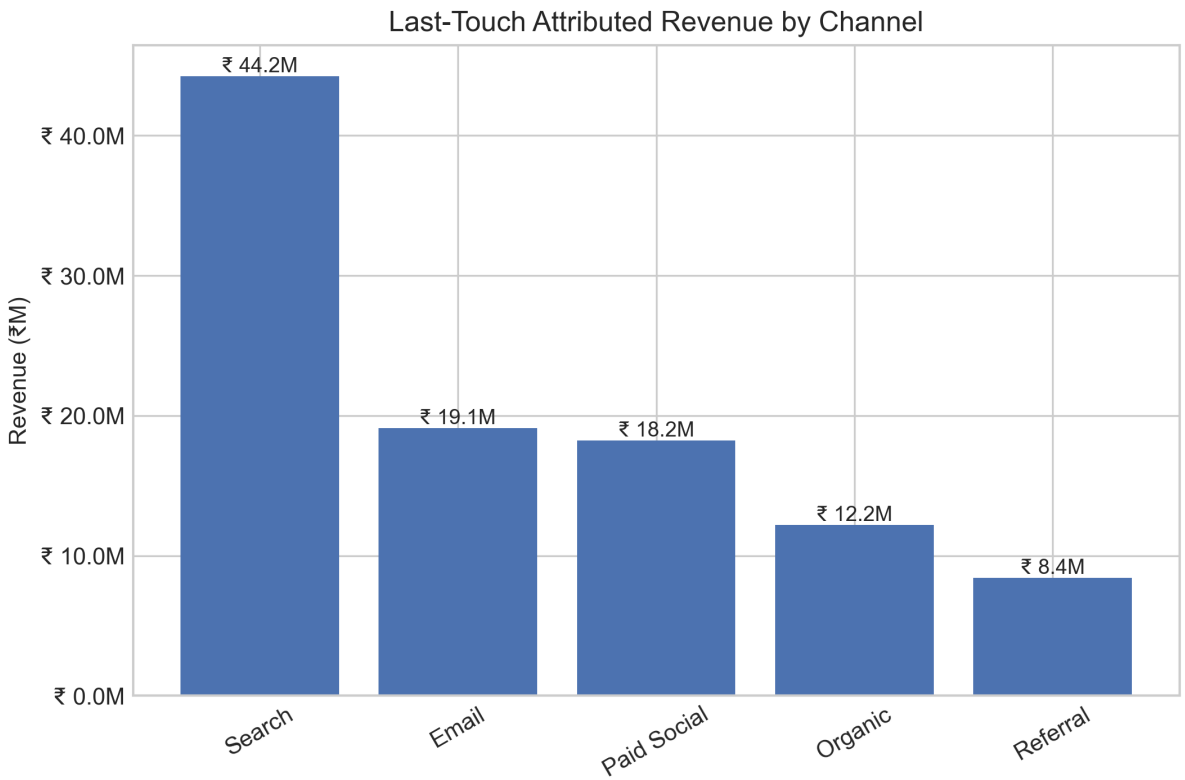

Figure 4: **Exhibit 4 — Search drives attributed revenue while Paid Social shows incremental impact**

Attribution highlights conversion channels while regression reveals true incremental drivers.

This indicates that relying only on attribution could undervalue channels that assist conversions earlier in the funnel.

Model Interpretation

- Attribution identifies conversion drivers
- Regression identifies incremental drivers
- Combined view provides stronger decision basis

Marketing Impact Modeling

Marketing spend shows measurable impact on revenue

Model R^2	0.58
MAPE	10%
MAE	Rs.80K

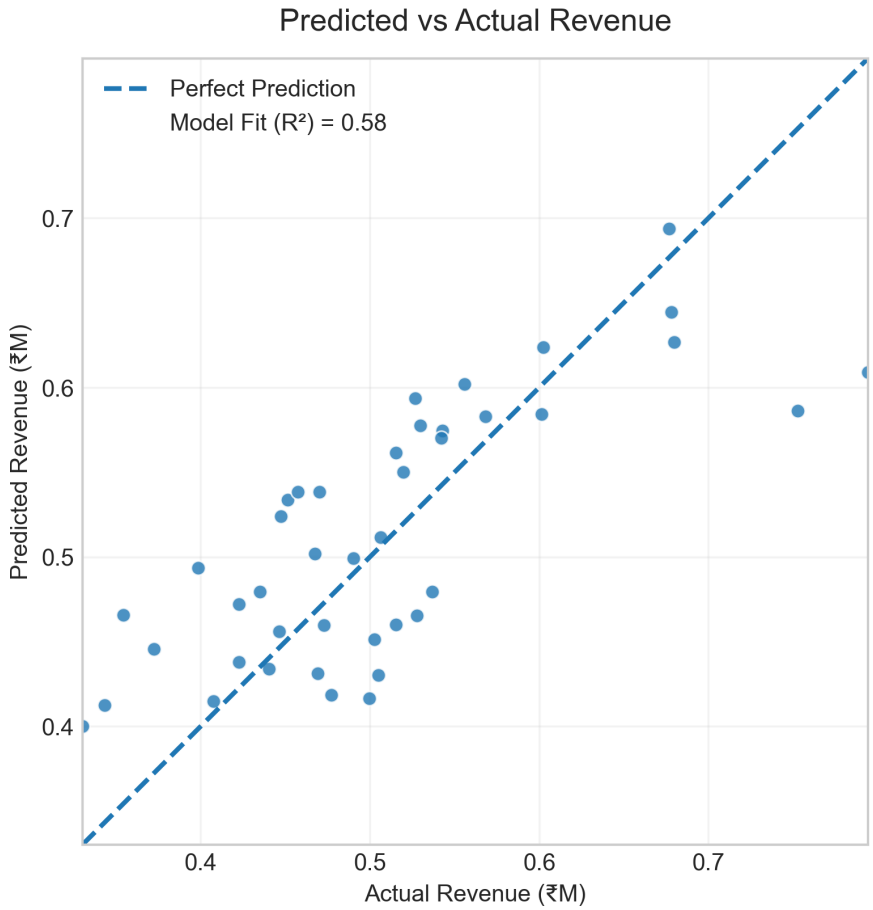

Figure 5: **Exhibit 5 — Model predictions closely track actual revenue**

Marketing spend shows strong positive relationship with revenue with diminishing returns.

This enables forward-looking marketing planning beyond attribution diagnostics to guide budget decisions.

Model Implications

- Marketing explains meaningful portion of revenue variation
- Predictive modeling enables forward planning
- Diminishing returns suggest careful scaling decisions

Budget Optimization Strategy

Optimized allocation improves efficiency without increasing spend

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Assumed next month marketing budget: **Rs.10.6M**

Constraints:

- Minimum allocation 5%
- Maximum allocation 40%

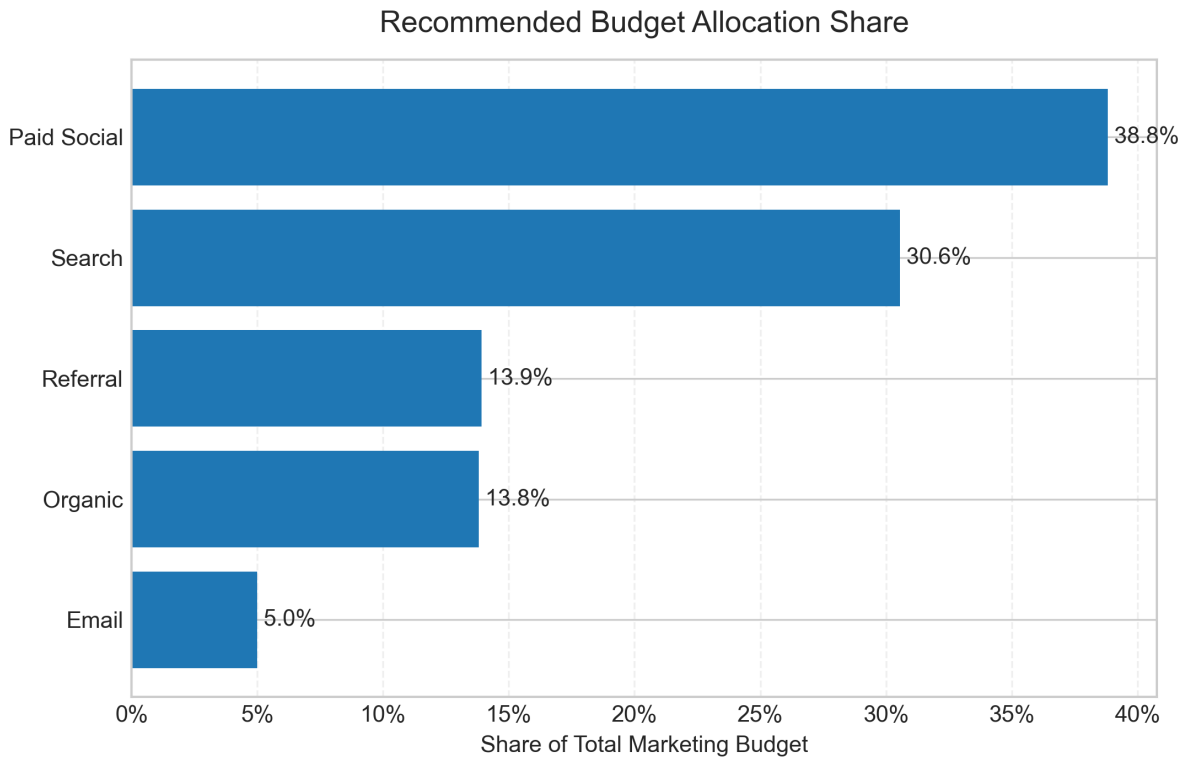


Figure 6: **Exhibit 6 — Optimized allocation improves expected efficiency**

The allocation reflects a balance between efficiency, scale potential, and diversification risk.

Allocation Rationale

- Increased allocation to high impact channels
- Controlled exposure to lower efficiency channels
- Diversification reduces concentration risk

Scenario Impact Analysis

Projected scenarios show stable expected revenue outcomes

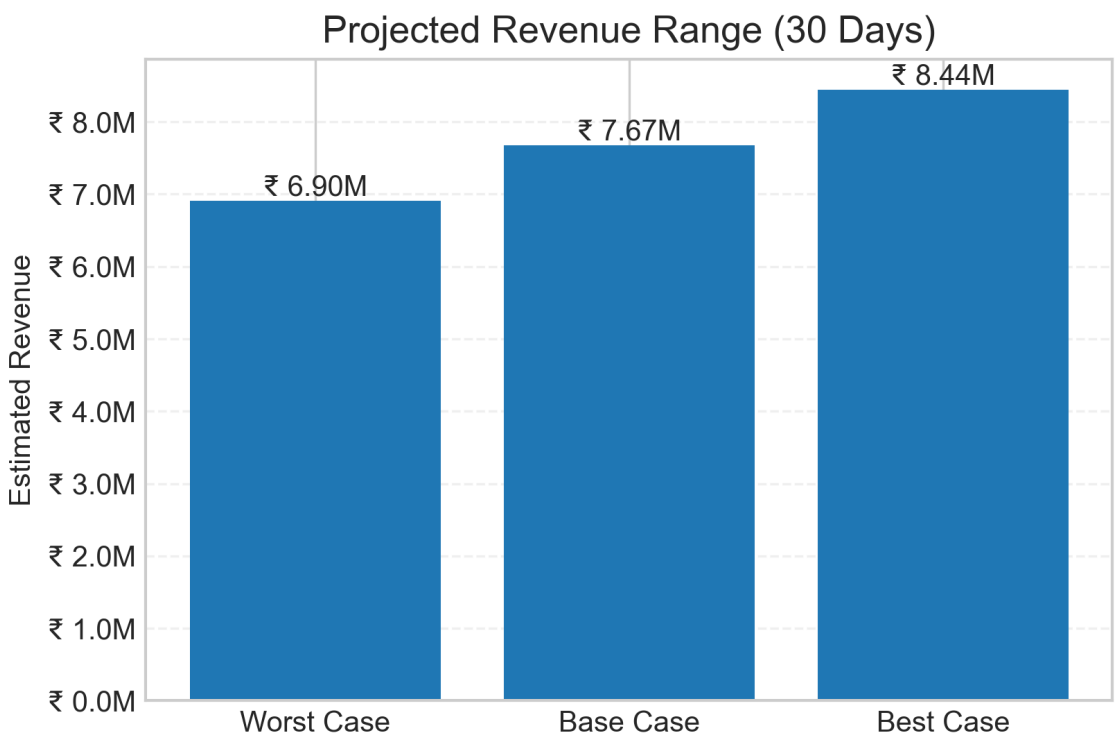


Figure 7: **Exhibit 7 — Projected revenue scenarios**

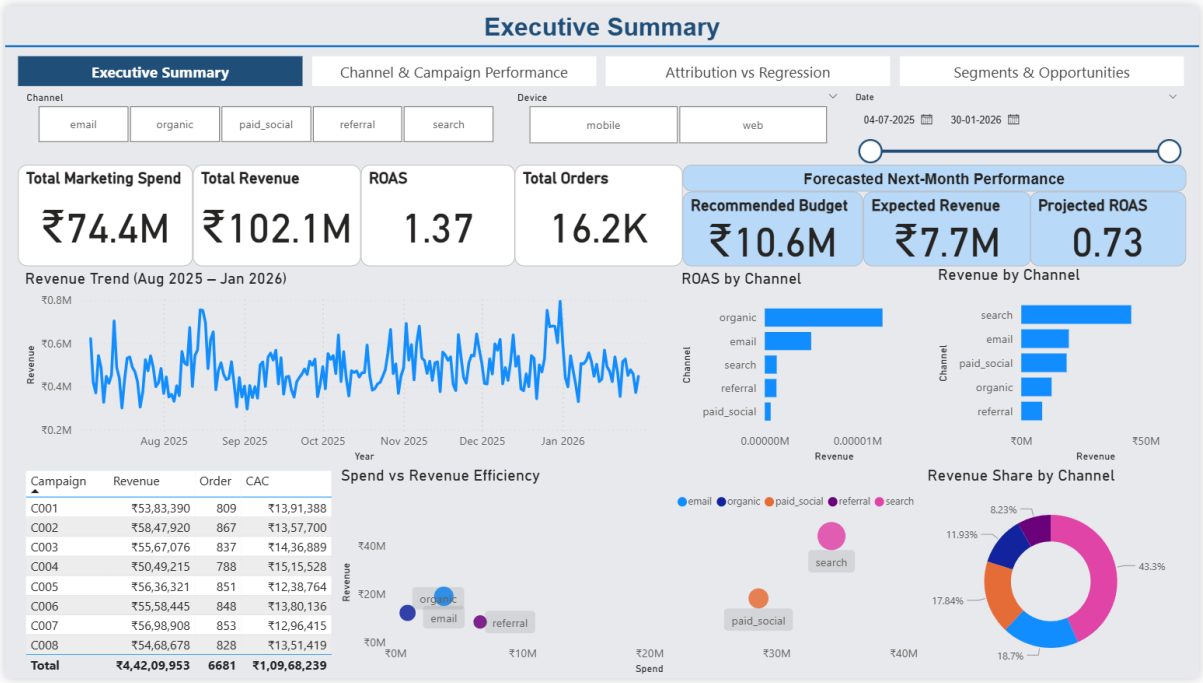
Worst Case	Rs.6.9M
Base Case	Rs.7.7M
Best Case	Rs.8.4M

Even under conservative assumptions, the optimized allocation maintains positive revenue efficiency.

Revenue Scenario Interpretation

- Revenue remains stable across scenarios
- Downside risk appears limited
- Upside driven by efficiency improvements

Executive Dashboard Views



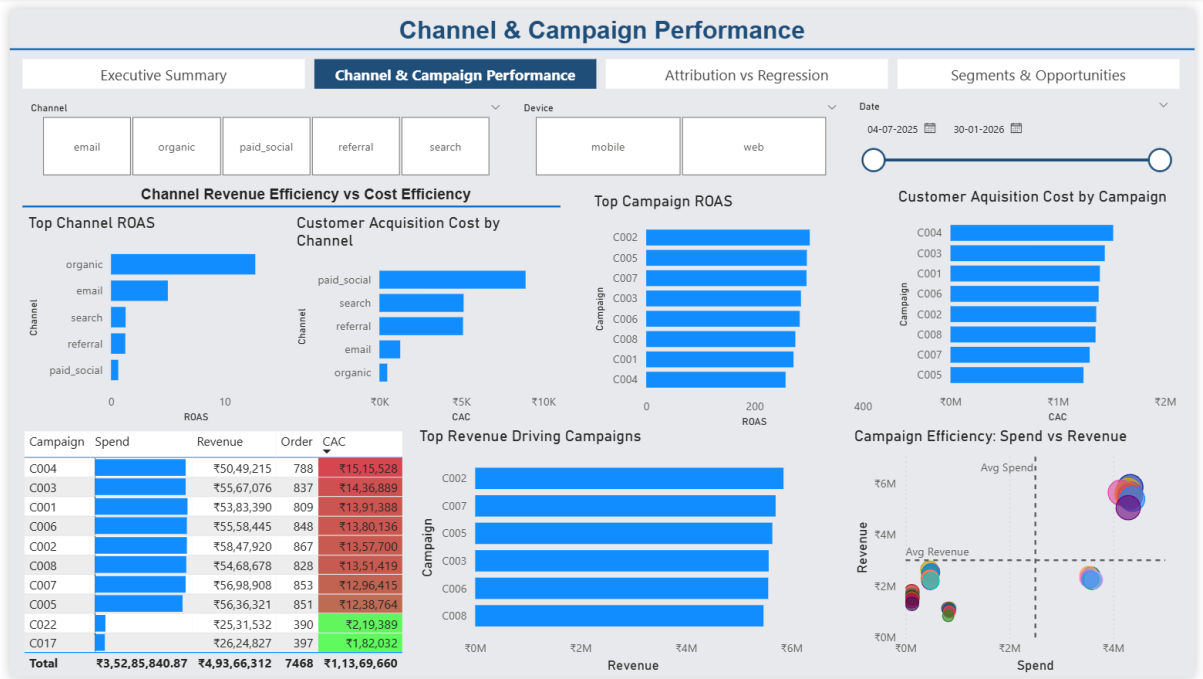


Figure 8: **Exhibit 8–9 — Executive and Campaign dashboards**

Channel efficiency differences highlight reallocation opportunity.

Paid Social shows highest CAC while Organic shows strongest efficiency.

Decision Summary

Channel	Diagnosis	Decision
Paid Social	Low ROI	Reduce spend
Search	High scale	Increase selectively
Organic	High ROAS	Increase investment
Email	Strong conversion	Maintain
Referral	Moderate	Maintain

Strategic Investment Thesis

Marketing investment should prioritize:

- High ROAS channels
- High marginal impact campaigns
- Retention growth
- High conversion segments

Expected Business Impact

If implemented:

- Improve marketing ROI
- Increase revenue efficiency
- Reduce inefficient spend
- Improve contribution margin
- Enable scalable decision framework

Skills Demonstrated

- Marketing Analytics
- Attribution Modeling
- Regression Modeling
- Budget Optimization
- Forecasting
- Power BI
- Business Strategy